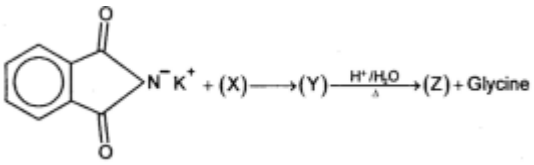
Quizrr Chapter-Wise Test for JEE Advanced - 2025

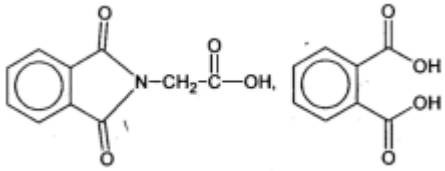
By: C I P H Ξ R

Q.1

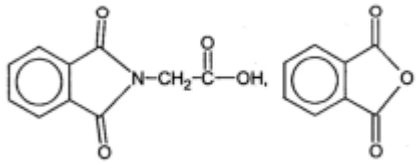


Compound X, Y, Z are:

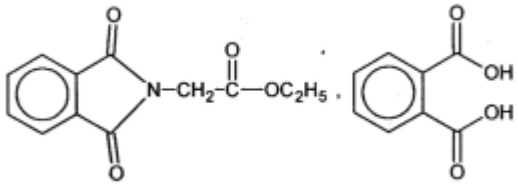
(A) Chloroacetic acid,



(B) Bromo acetic acid,

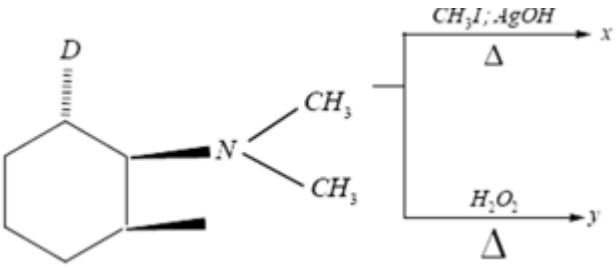


(C) Ethyl chloro acetic acid,



(D) None of the above

Q.2



x and y are

(A) Stereo Isomers

(B) Positional Isomers

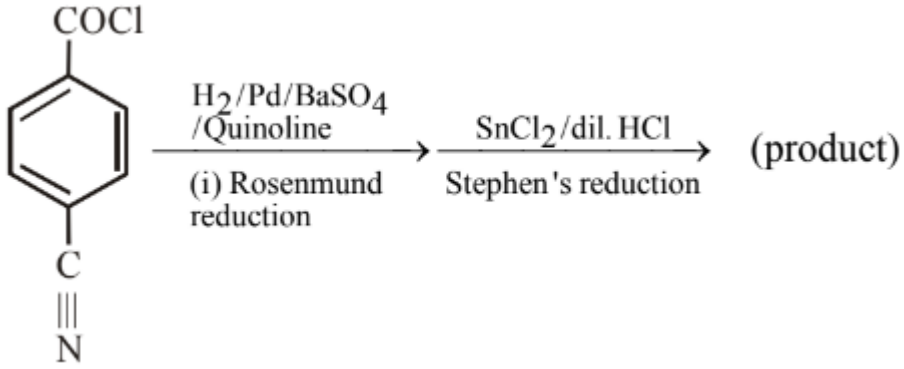
(C) Chain Isomers

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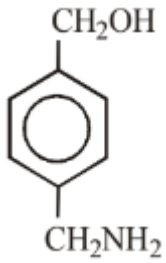
(D) Not Isomers

Q.3

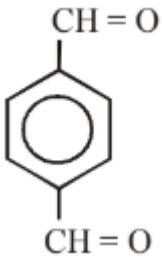
The product of following reaction is -



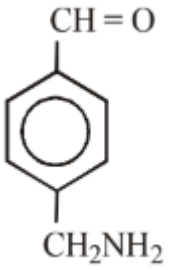
(A)



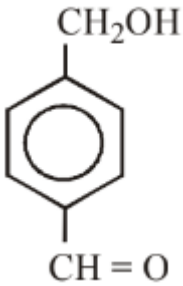
(B)



(C)

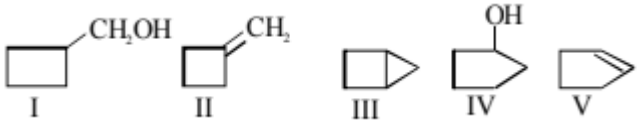


(D)



Q.4

Treatment of cyclobutyl methyl amine with Nitrous Acid does not give



(A) I, II, IV

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(B) II, V, I

(C) I, II, IV, V

(D) III only

Q.5

Consider the following reaction,

$$C_6H_5NH_2 + CHCl_3 + KOH \xrightarrow{\Delta} (A) \xrightarrow{H^+/H_2O} (B) + (C)$$

The compounds (B) and (C) are:

(A) C_6H_5COOH and NH_3 respectively

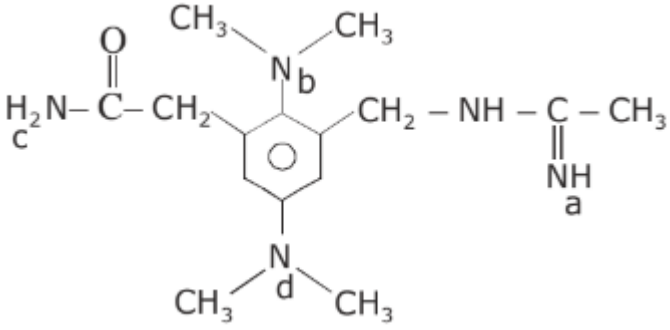
(B) $C_6H_5NH_2$ and $HCOOH$ respectively

(C) $C_6H_5NH_2$ and H_2O respectively

(D) None of these.

Q.6

Basicity order in following compound is :



(A) b > d > a > c

(B) a > b > d > c

(C) a > b > c > d

(D) a > c > b > d

Q.7

Identify (A) and (B) in the given reaction, $CH_3CH_2NH_2 + (CH_3CO)_2O \xrightarrow{\Delta} (A) + (B)$

(A) $CH_3CH_2OH, CH_3CONHCH_2CH_3$

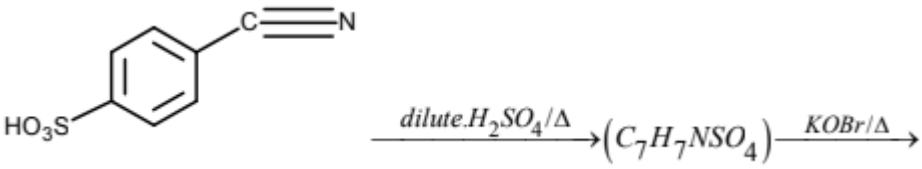
(B) $CH_3COOH, CH_3CH_2NHCOCH_3$

(C) $CH_3CH_2COOH, CH_3NHCOCH_3$

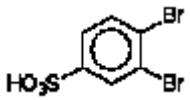
(D) $CH_3COOH, CH_3CONHCH_3$

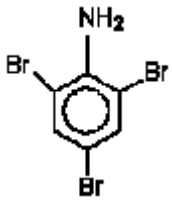
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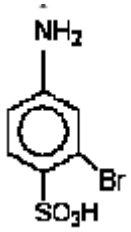
Q.8

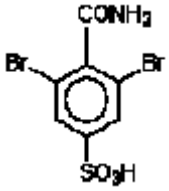


Product $\xrightarrow{Br_2/H_2O}$ end product. The end product is

(A) 

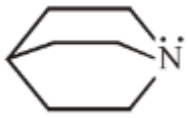
(B) 

(C) 

(D) 

Q.9

Which of the following statement is/are correct —

(A) 

is a stronger nucleophile than triethylamine

(B) $H_2\ddot{N}-\ddot{O}H$

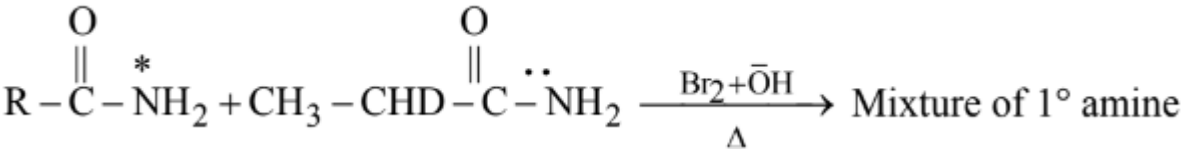
is stronger nucleophile than $\ddot{N}H_3$

(C) $R-\ddot{S}H$ is a stronger nucleophile than $R-\ddot{O}H$ is polar protic solvent like ethanol

(D) OH^- is stronger nucleophile than H_2O

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Q.10



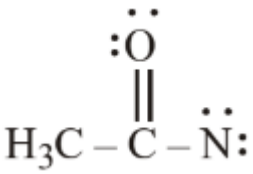
Which of the following statements is/are correct for above reaction

(A) A mixture of four different amines are formed confirming that a free carbonylnitrene intermediate is formed.

(B) A mixture of two amines are formed confiring that rearrangement is strictly intramolecular

(C) If R is chiral, it migrates with retention of configuration.

(D) A free carbonylnitrene



intermediate is formed and only two amines are formed.

Q.11

Which of the following reactions give only primary amine as a product?

(A) Hofmann's bromamide reaction

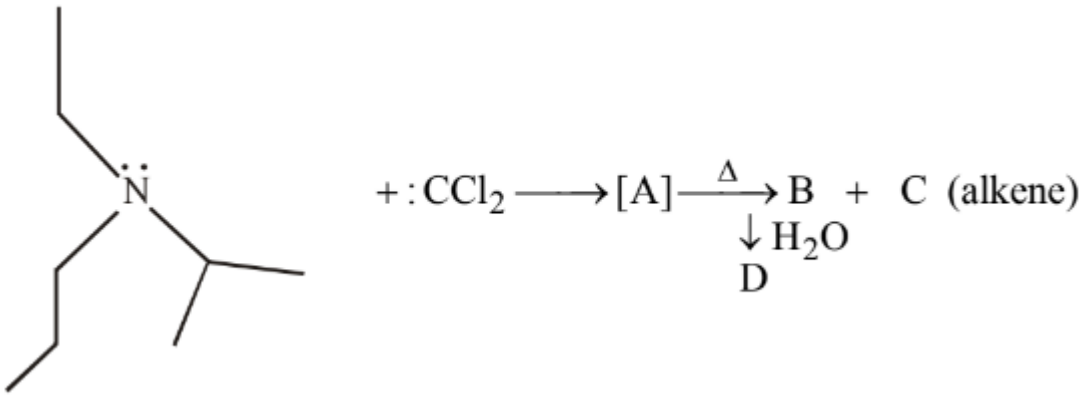
(B) Reduction of oximes with $LiAlH_4$

(C) Reaction of alcohol with NH_3 at high temperature in presence of alumina

(D) Reduction of nitro compounds with Sn and HCl

Q.12

Passage:

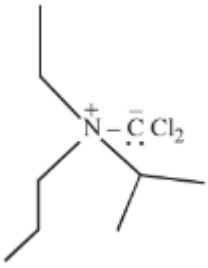


Question:

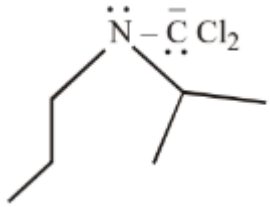
The compound B is –

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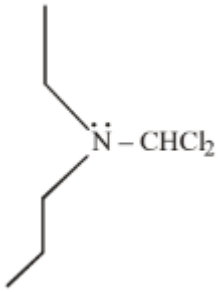
(A)



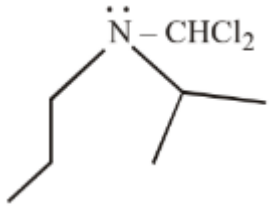
(B)



(C)

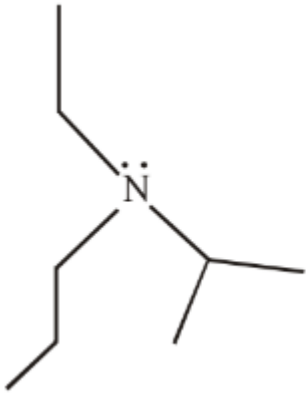


(D)



Q.13

Passage:



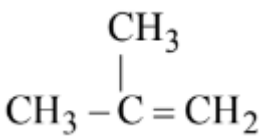
$$+ : CCl_2 \longrightarrow [A] \xrightarrow{\Delta} B + C \text{ (alkene)}$$
$$\downarrow H_2O$$
$$D$$

Question:

The compound C is –

(A) $CH_2 = CH_2$

(B)



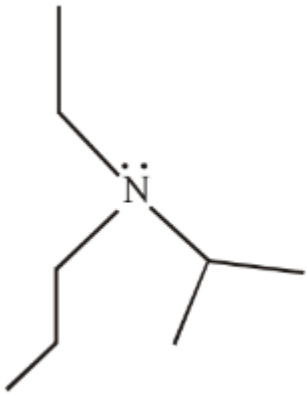
(C) $CH_2 = CCl_2$

(D) $CH_3 - CH = CH - CH_3$

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Q.14

Passage:



$$+ : CCl_2 \longrightarrow [A] \xrightarrow{\Delta} B + C \text{ (alkene)}$$
$$\downarrow H_2O$$
$$D$$

Question:

The compound D is -

(A) N-isopropyl-N-propyl ethanamide

(B) N-ethyl-N-isopropyl-ethanamine

(C) N-isopropyl-N-propyl methanamine

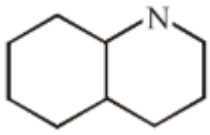
(D) N-isopropyl-N-propyl methanamide

Q.15

Passage:

An organic compound A has molecular formula $C_9H_{13}NO$ and it can be resolved into enantiomers. A does not decolourise bromine water solution. A on refluxing with dilute H_2SO_4 solution yields another resolvable compound, B $(C_9H_{14}O_3)$ which gives effervescence with $NaHCO_3$.B on treatment with $NaBH_4$ yields C $(C_9H_{16}O_3)$ .C on heating with concentrated H_2SO_4 solution yields a sweet smelling neutral compound D $(C_9H_{14}O_2)$.

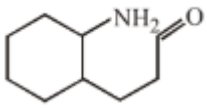
Compound A on reduction with $LiAlH_4$ yields E $(C_9H_{19}O)$ also which on treatment with H_2SO_4 yields the following compound



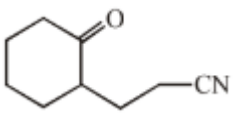
Question:

Organic compound A is —

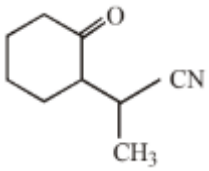
(A)

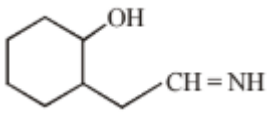


(B)



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(C) 

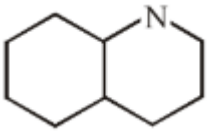
(D) 

Q.16

Passage:

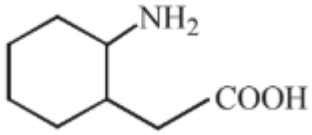
An organic compound A has molecular formula $C_9H_{13}NO$ and it can be resolved into enantiomers. A does not decolourise bromine water solution. A on refluxing with dilute H_2SO_4 solution yields another resolvable compound, B $(C_9H_{14}O_3)$ which gives effervescence with $NaHCO_3$.B on treatment with $NaBH_4$ yields C $(C_9H_{16}O_3)$ .C on heating with concentrated H_2SO_4 solution yields a sweet smelling neutral compound D $(C_9H_{14}O_2)$.

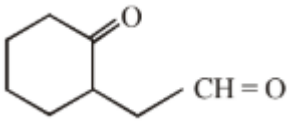
Compound A on reduction with $LiAlH_4$ yields E $(C_9H_{19}O)$ also which on treatment with H_2SO_4 yields the following compound

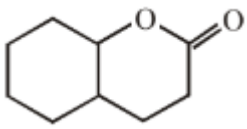


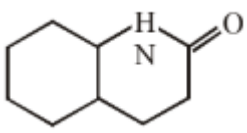
Question:

The sweet smelling neutral compound D $(C_9H_{14}O_2)$ is -

(A) 

(B) 

(C) 

(D) 

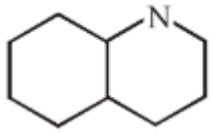
Q.17

Passage:

An organic compound A has molecular formula $C_9H_{13}NO$ and it can be resolved into enantiomers. A does not decolourise bromine water solution. A on refluxing with dilute H_2SO_4 solution yields another resolvable compound, B $(C_9H_{14}O_3)$ which gives effervescence with $NaHCO_3$.B on treatment with $NaBH_4$ yields C $(C_9H_{16}O_3)$ .C on heating with concentrated H_2SO_4 solution yields a sweet smelling neutral compound D $(C_9H_{14}O_2)$.

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Compound A on reduction with $LiAlH_4$ yields E $(C_9H_{19}O)$ also which on treatment with H_2SO_4 yields the following compound



Question:

Due to reduction of optically pure B two isomeric products C form. Isomeric products C are -

(A) enantiomers

(B) diastereomers

(C) position isomers

(D) functional isomers

Q.18

Match the column:

Column I	**Column II**
(A) $R-CN \longrightarrow R-CH_2-NH_2$	(p) B_2H_6, THF
(B) $R-C\equiv CH \longrightarrow R-CH_2-CHO$	(q) Sia_2BH, $H_2O_2^-$, OH^-
(C) $R-CO-N(R_2)(R_2) \longrightarrow R-CHO$	(r) $LiAl(OC_2H_5)_3H$
(D) $R-CHO \longrightarrow R-CH_2OH$	(s) $LiAlH_4$
	(t) $NaBH_4$

(A) (A) - p,q, (B) - r, (C) - r,s, (D) - p,t

(B) (A) - q,r, (B) - q,s, (C) - r,t, (D) - s,p,t

(C) (A)-p,s, (B)-q, (C)-r, (D)-s,t

(D) (A) - p, (B) - r, (C) - p,t, (D) - p,t

Q.19

Two isomeric compounds (A) and (B) of molecular formula C_4H_7N do not react with HNO_2 or acetyl chloride. On refluxing separately with dil. HCl, (A) gives (C) whereas (B) gives (D) as products. Both (C) and (D) are isomeric compounds which react with $NaOH$. What is the volume in ml of $5/3$ $NNaOH$ that react with 0.88 g of (C) or (D).

Q.20

To carry out the reduction of $Ph-NO_2$ to $Ph-NH_2$, how many reagents among the following can be successfully used?

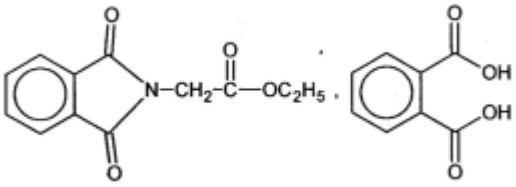
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$Fe/H_2O, Sn/HCl, H_2/Pd - C, NH_4HS, Zn - NH_4Cl/H_2O, Na_3AsO_3, LiAlH_4, SnCl_2/HCl, Fe/HCl, Na_2S_n, LiAlH_4 - TiCl_4$

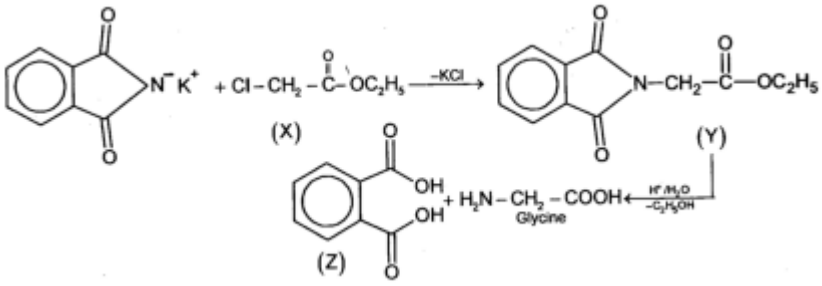
Answers & Solutions

Q.1 Answer:

Ethyl chloro acetic acid,



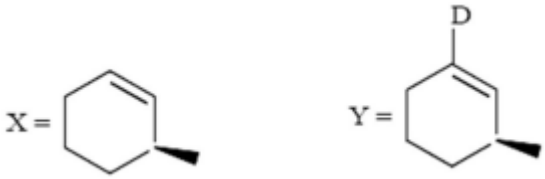
Solution:



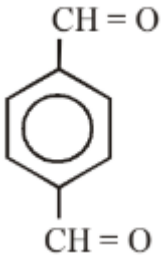
Q.2 Answer:

Not Isomers

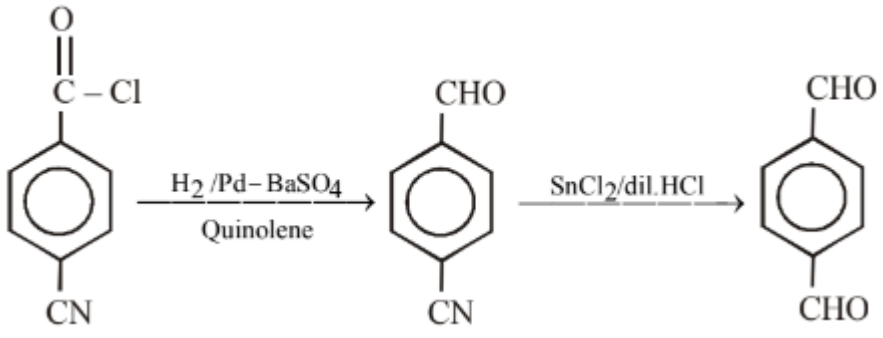
Solution:



Q.3 Answer:



Solution:

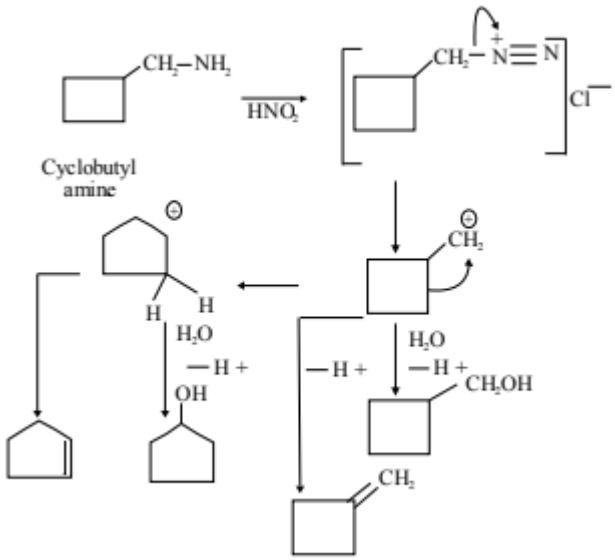


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Q.4 Answer:

III only

Solution:



Q.5 Answer:

$C_6H_5NH_2$ and $HCOOH$ respectively

Solution:

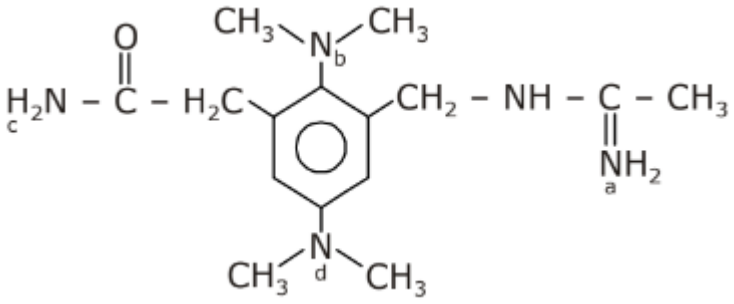
$$C_6H_5NH_2 + CHCl_3 + 3KOH \longrightarrow \underset{(A)}{C_6H_5N \equiv C} + 3KI + 3H_2O$$

$$\underset{(A)}{C_6H_5N \equiv C} + 2H_2O \xrightarrow{H^+} \underset{(B)}{C_6H_5NH_2} + \underset{(C)}{HCOOH}$$

Q.6 Answer:

a > b > d > c

Solution:



a is most basic due to high charge density.

b is more basic than d and c due to ortho effect.

c is least basic because I.p. of nitrogen is conjugated with $-\overset{\overset{O}{\|}}{C}-$ **group.**

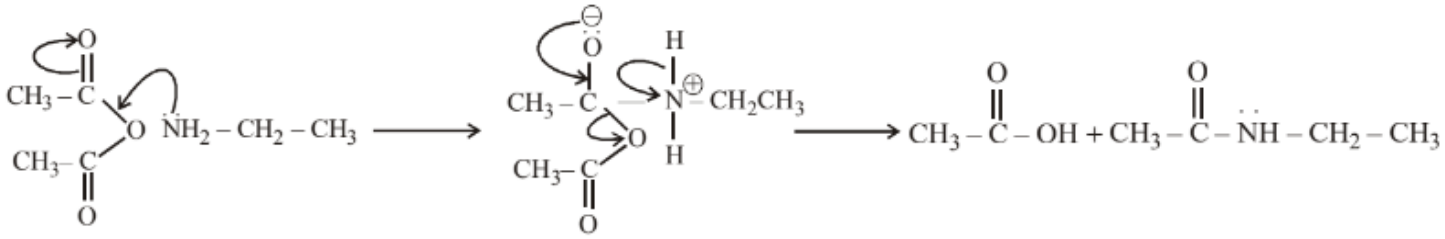
Order : $a > b > d > c$

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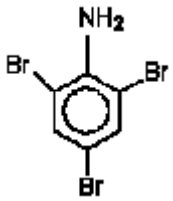
Q.7 Answer:

$CH_3COOH, CH_3CH_2NHCOCH_3$

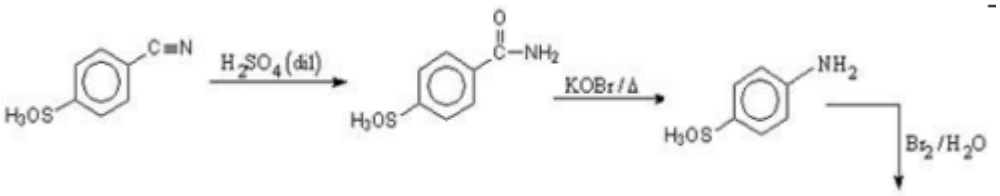
Solution:

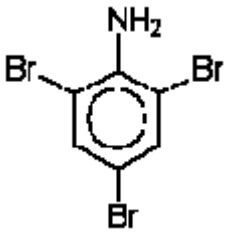


Q.8 Answer:

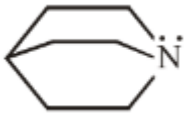


Solution:



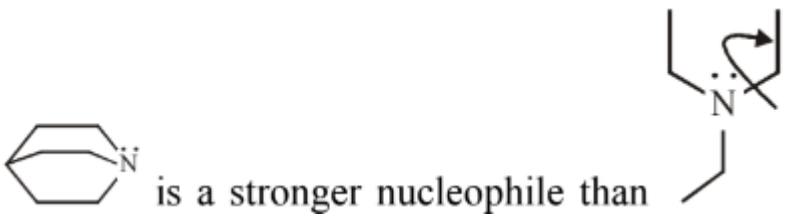


Q.9 Answer:



is a stronger nucleophile than triethylamine

Solution:



is a stronger nucleophile than

because in former all β-carbon are tied up with a carbon while in latter the substituent bonded to nitrogen

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are almost free to rotate causes seteric hindrance during approach of unshared pair of electron to the attacking site.

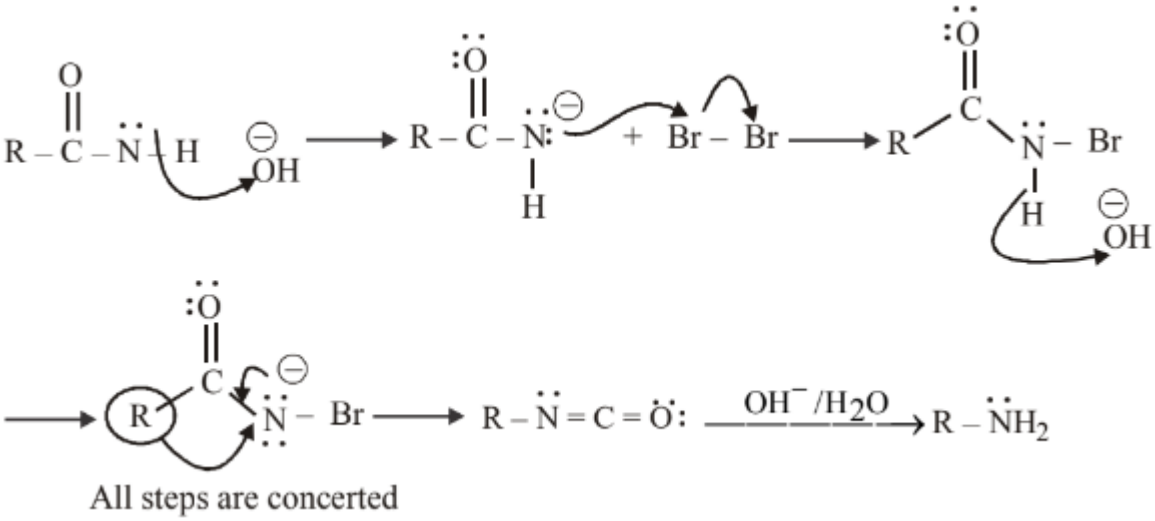
$R-\ddot{S}H$ is stronger nucleophile than $R-\ddot{O}H$

due to less salvation in polar protic solvent like ethanol.

Q.10 Answer:

A mixture of two amines are formed confiring that rearrangement is strictly intramolecular

Solution:



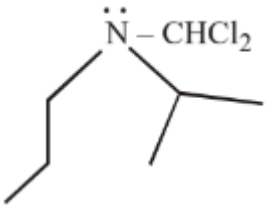
Q.11 Answer:

Hofmann's bromamide reaction

Solution:

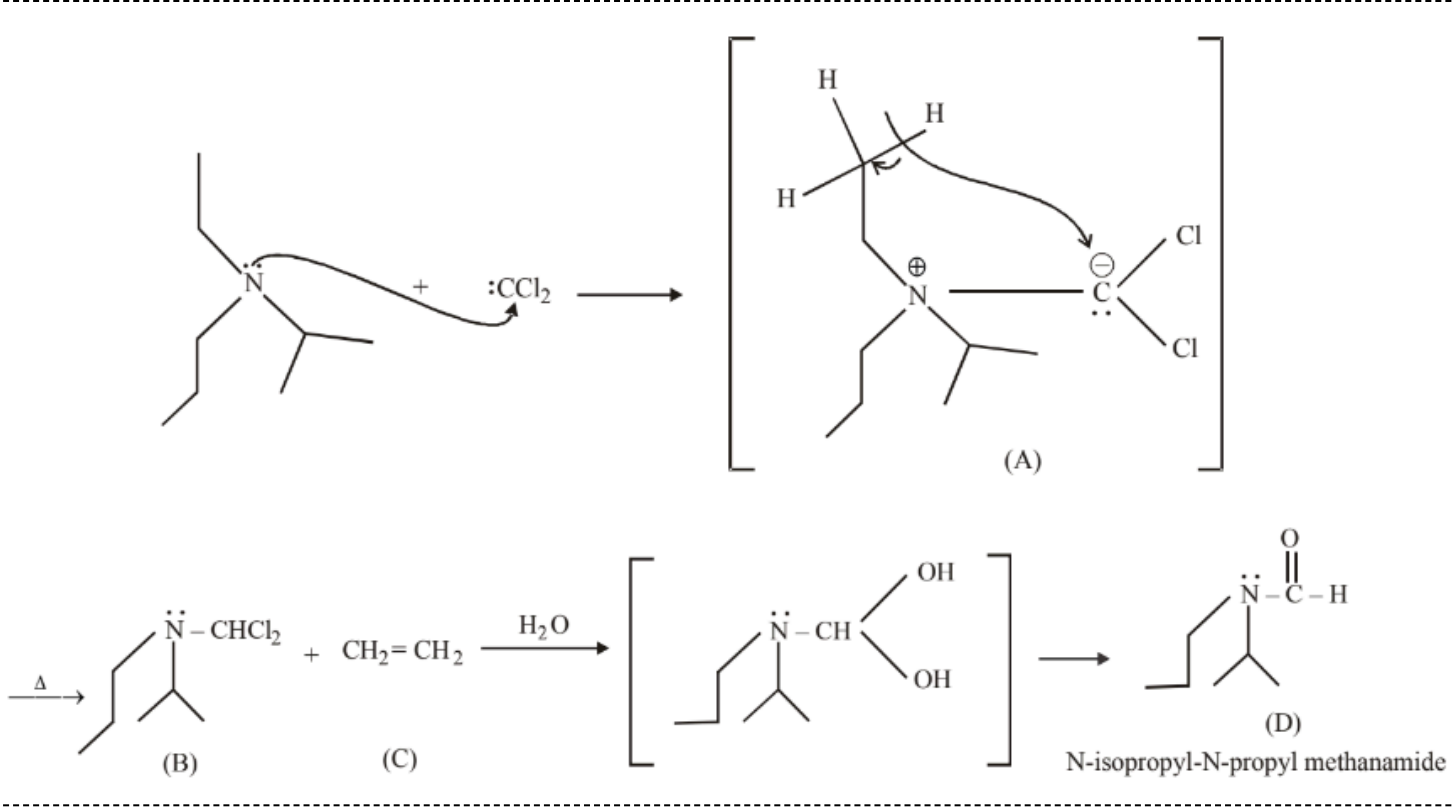
When excess alcohol vapours are mixed with ammonia and passed over heated alumina mixture of $1^\circ, 2^\circ$ and 3° amines are formed. All other methods (A), (B) \& (D) give only 1° amines.

Q.12 Answer:



Solution:

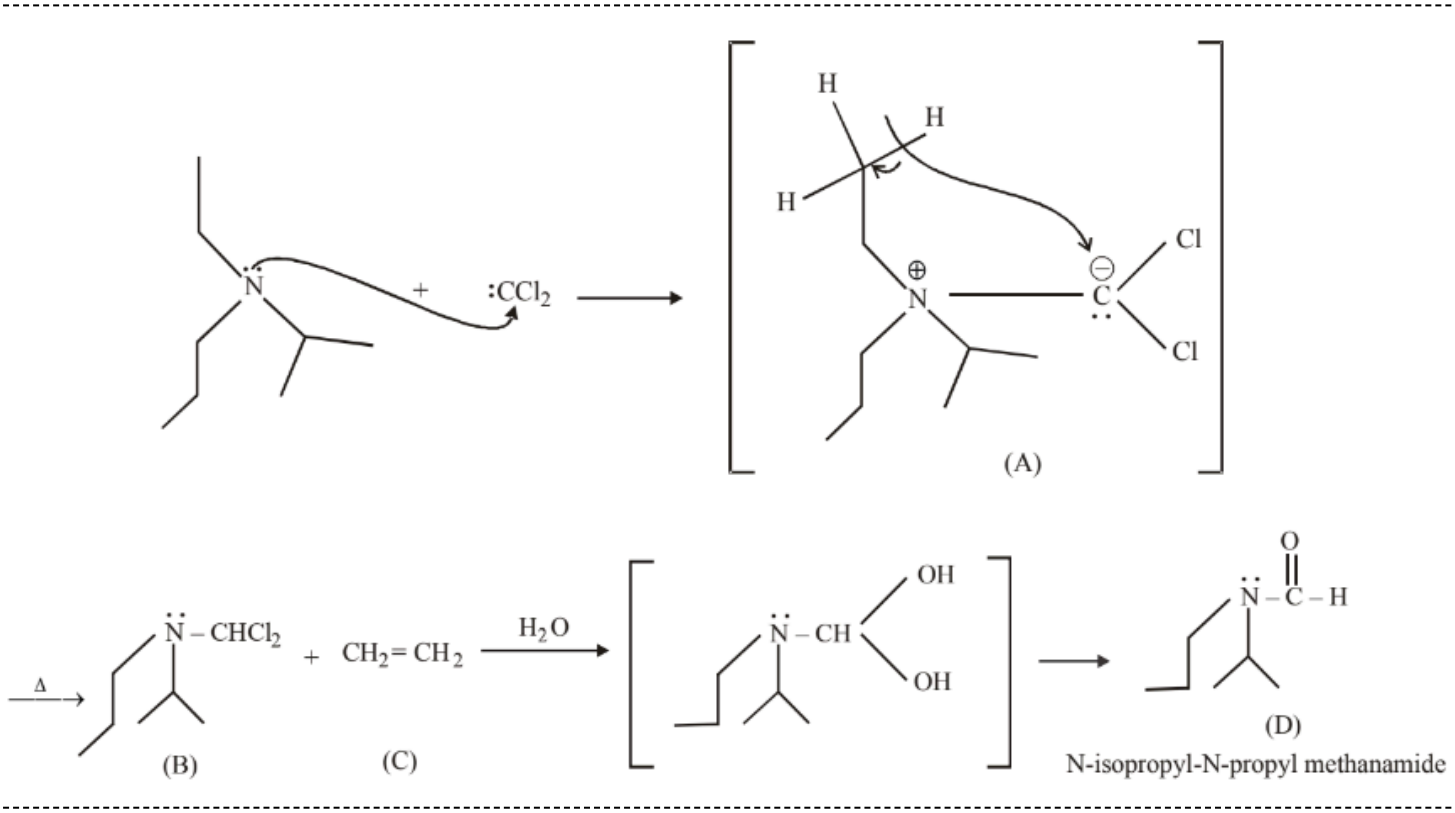
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Q.13 Answer:

$CH_2 = CH_2$

Solution:

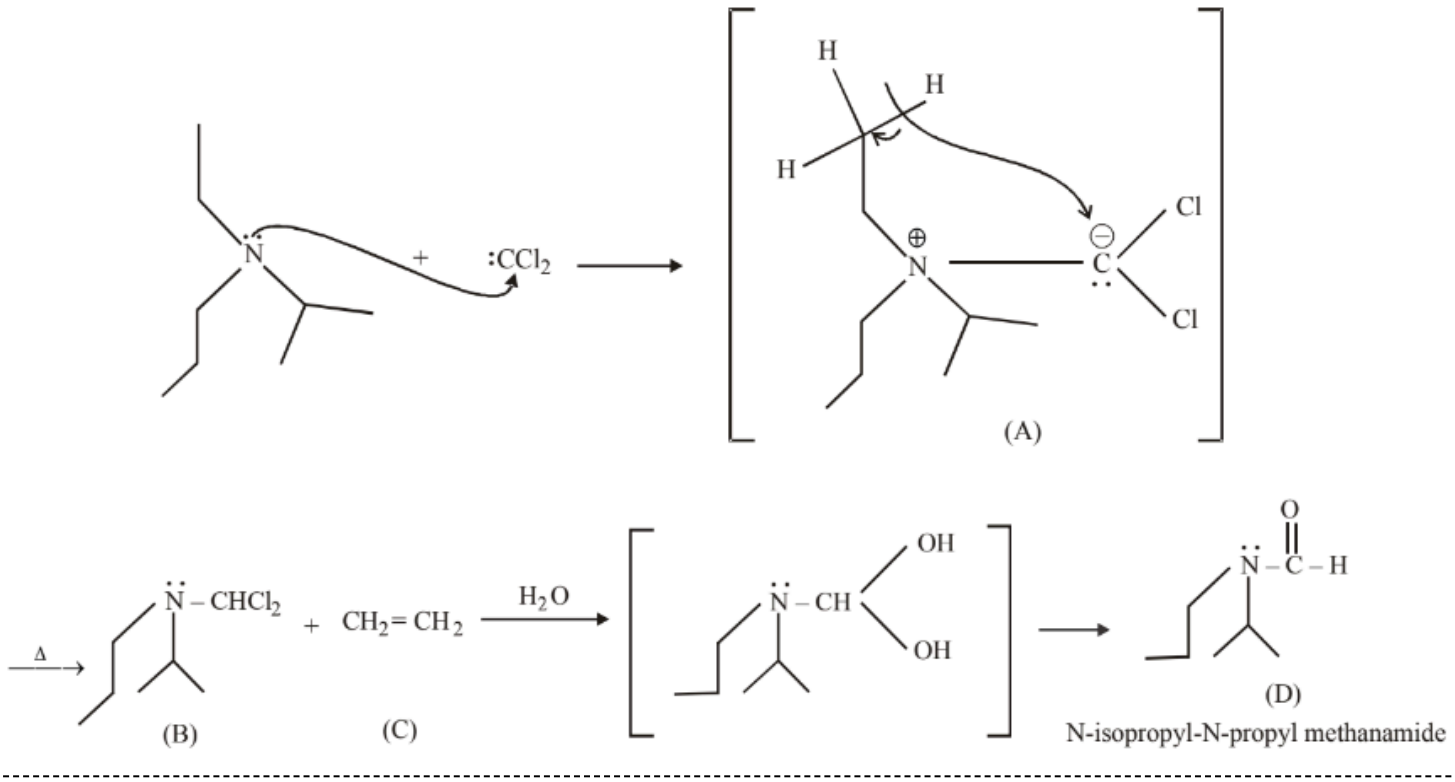


Q.14 Answer:

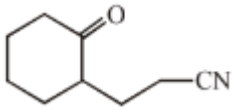
N-isopropyl-N-propyl methanamide

Solution:

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Q.15 Answer:



Solution:

A

$C_9H_{12}NO$

Optically active

does not decolourise bromine water

$\Rightarrow$ no unsaturation

$\xrightarrow[H_2SO_4]{\text{Refluxed with}}$

Resolvable compound

B

$C_9H_{14}O_3$

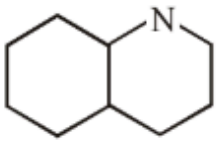
(Nitrogen is lost, 2 oxygens gained)

effervescence with $NaHCO_3$

$\Rightarrow$ COOH group is present

$\downarrow LiAlH_4$

$\downarrow H_2SO_4$



$\downarrow NaBH_4$

C

$(C_9H_{16}O_3)$

$\downarrow$ Conc. H_2SO_4, Δ

D $(C_9H_{14}O_2)$

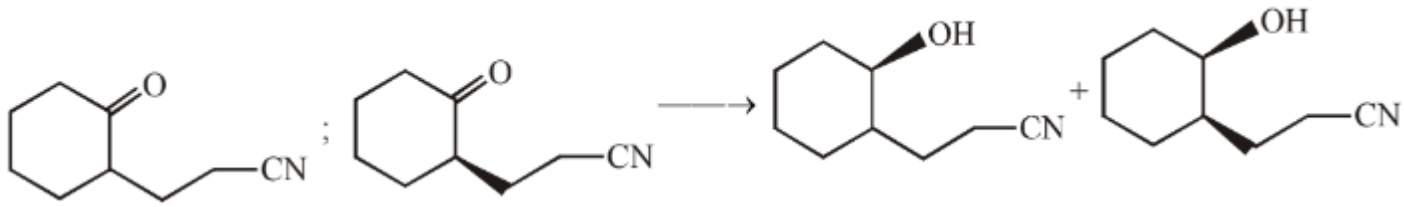
Sweet smelling

$\Rightarrow$ Ester

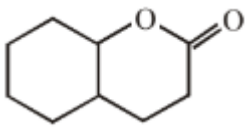
If D is ester then C has $-OH$ and COOH groups which is confirmed by B also. Thus, we can say that A has $-CN$ group.

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Comparing with final product we can say A is



Q.16 Answer:



Solution:

A

$C_9H_{12}NO$

Optically active

does not decolourise bromine water

$\Rightarrow$ no unsaturation

$\xrightarrow[H_2SO_4]{\text{Refluxed with}}$

Resolvable compound

B

$C_9H_{14}O_3$

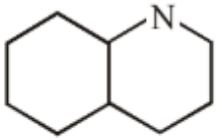
(Nitrogen is lost, 2 oxygens gained)

effervescence with $NaHCO_3$

$\Rightarrow$ COOH group is present

$\downarrow LiAlH_4$

$\downarrow H_2SO_4$



$\downarrow NaBH_4$

C

$(C_9H_{16}O_3)$

$\downarrow$ Conc. H_2SO_4, Δ

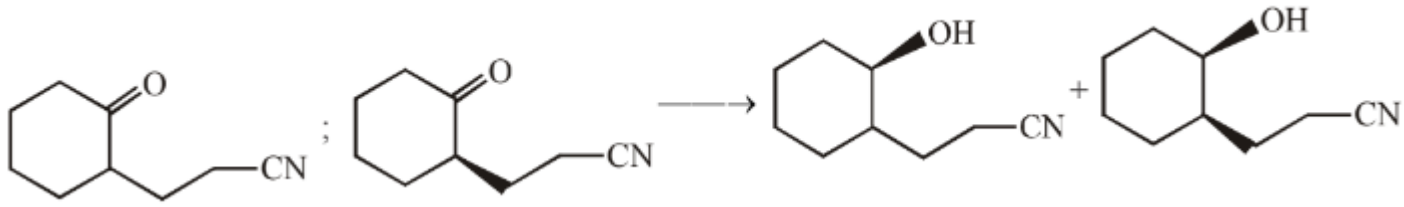
D $(C_9H_{14}O_2)$

Sweet smelling

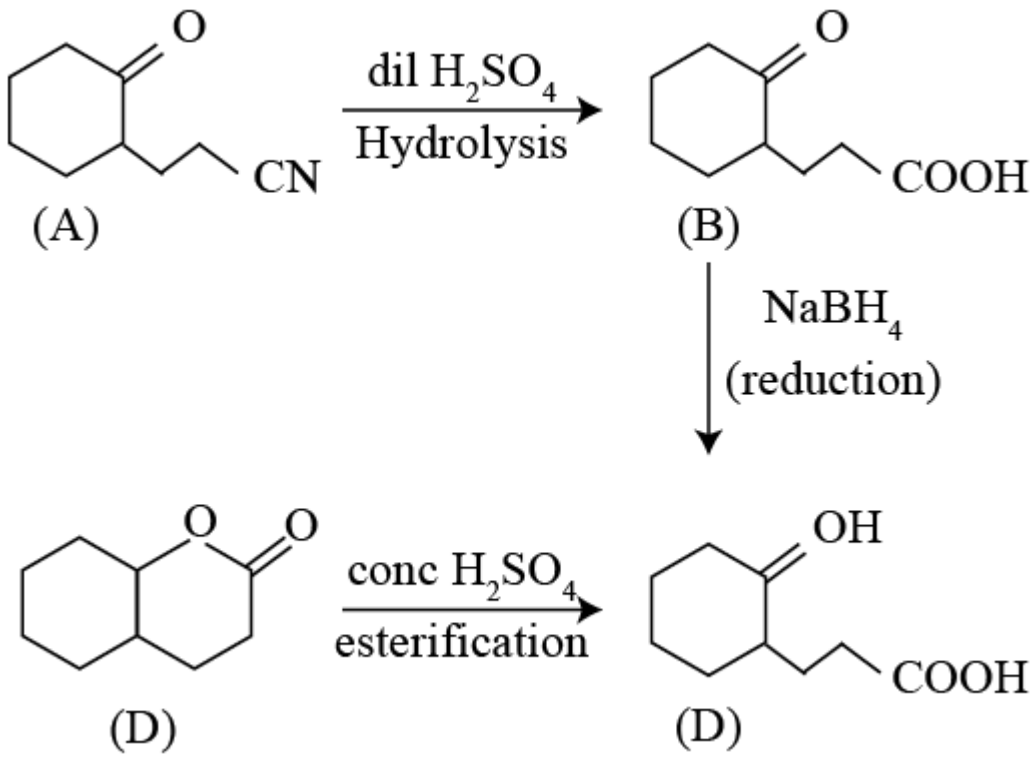
$\Rightarrow$ Ester

If D is ester then C has $-OH$ and COOH groups which is confirmed by B also. Thus, we can say that A has $-CN$ group.

Comparing with final product we can say A is



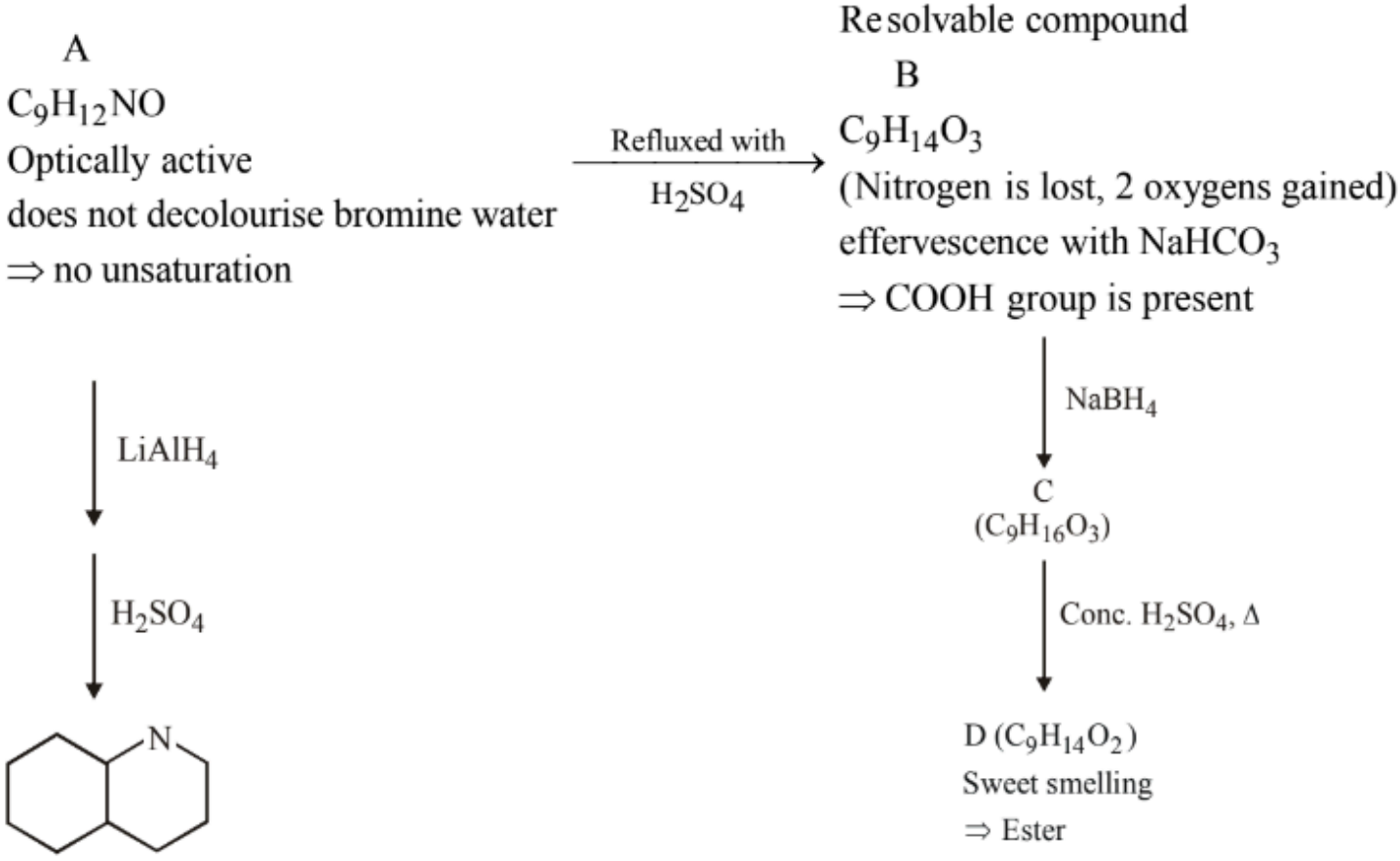
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Q.17 Answer:

diastereomers

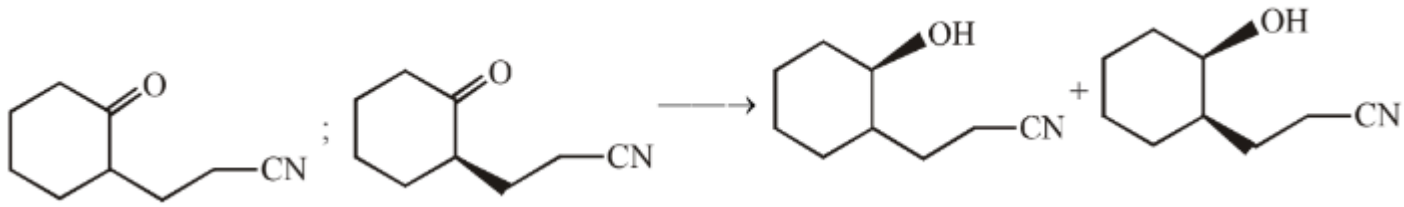
Solution:



If D is ester then C has $-OH$ and COOH groups which is confirmed by B also. Thus, we can say that A has $-CN$ group.

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Comparing with final product we can say A is



Q.18 Answer:

(A)-p,s, (B)-q, (C)-r, (D)-s,t

Solution:

$(\mathbf{A}) - \mathbf{ps}, (\mathbf{B}) - \mathbf{q}, (\mathbf{C}) - \mathbf{r}, (\mathbf{D}) - \mathbf{st}$

$R - CN \xrightarrow{B_2H_6, THF} R - CH_2 - NH_2$

$R - CN \xrightarrow{LiAlH_4} R - CH_2 - NH_2$

$$R - C \equiv C - H \xrightarrow[H_2O_2^-, OH^-]{Sia_2BH} R - CH_2 - CHO$$

$$R_1 - CO - \underset{\displaystyle R_2}{\underset{|}{N}} - R_2 \xrightarrow{LiAl(OEt)_3H} R - CHO$$

$R - CHO \xrightarrow{LiAlH_4} R - CH_2 - OH; R - CHO \xrightarrow{NaBH_4} R - CH_2 - OH$

Q.19 Answer:

—

Solution:

Since A and B do not react with HNO_2 and acetyl chloride they could not be amines.

$$(A)\ \ CH_3 - CH_2 - CH_2 - CN \xrightarrow{H_3O^+} CH_3\,CH_2\,CH_2\,COOH\ (C)$$

$$(B)\ \ \underset{\displaystyle CH_3}{\underset{|}{CH_3CH}}CN \xrightarrow{H_3O^+} \underset{\displaystyle CH_3}{\underset{|}{CH_3CH}} - COOH\ (D)$$

Molecular weight of C or D $= 88$ g

$\therefore \frac{0.88}{88} = \frac{V}{1000} \times \frac{5}{3}; V = 6ml$

Q.20 Answer:

—

Solution:

Only suitable reagents for the conversion of $PhNO_2$ to $PhNH_2$ are

$Sn/HCl, H_2/Pd - C, NH_4HS, SnCl_2/HCl, Fe/HCl$ **and** $Na_2\,S_n$.

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